

TUTORIAL 6: ORDINARY DIFFERENTIAL EQUATIONS

- 1 Solve the following ODE using separable variables.
 - a. $\frac{dy}{dx} = \frac{\sec^2 y}{1+x^2}$
 - b. $x \frac{dv}{dx} = \frac{1-4v^2}{3v}$
 - c. $\frac{dx}{dt} + x^2 = x$
 - d. $\frac{dy}{dx} = 3x^2(1+y^2)$
 - e. $y^{-1}dy + ye^{\cos x} \sin x dx = 0$
 - f. $(x+xy^2)dx + e^{x^2}ydy = 0$
- 2 Solve the following Initial Value Problem.
 - a. $y' = x^3(1-y), \quad y(0) = 3$
 - b. $\frac{dy}{dx} = (1+y^2) \tan x, \quad y(0) = \sqrt{3}$
 - c. $\frac{dy}{dx} = 2\sqrt{y+1}(\cos x), \quad y(\pi) = 0$
 - d. $\frac{dy}{dx} = 2x \cos^2 y, \quad y(0) = \frac{\pi}{4}$
- 3 Obtain the general solution to the following ODE.
 - a. $\frac{dr}{d\theta} + r \tan \theta = \sec \theta$
 - b. $(t+y+1)dt - dy = 0$
 - c. $(x^2+1)\frac{dy}{dx} + xy = x$
 - d. $(x^2+1)\frac{dy}{dx} = x^2 + 2x - 1 - 4xy$
- 4 Solve the following Initial Value Problem.
 - a. $\frac{dy}{dx} + 4y - e^{-x} = 0, \quad y(1) = e - 1$
 - b. $\frac{dy}{dx} + 4y - e^{-x} = 0, \quad y(0) = \frac{4}{3}$
 - c. $\frac{dy}{dx} + \frac{3y}{x} + 2 = 3x, \quad y(1) = 1$
 - d. $\sin x \frac{dy}{dx} + y \cos x = x \sin x, \quad y(\frac{\pi}{2}) = 2$
- 5 **Newton's Law of Cooling.** According to Newton's Law of Cooling, if an object at temperature T is immersed in a medium having the constant temperature M , then the rate of change of T is proportional to the difference of temperature $M - T$. This gives the differential equation,

$$\frac{dT}{dt} = k(M - T)$$
 - (a) Solve the equation for T .
 - (b) A thermometer reading 100° is placed in a medium having the constant temperature of 70° . After 6 minutes, the thermometer reads 80° . What is the reading after 20 min?

- 6 **Electric Circuit.** The simplest electric circuit shown in Figure 1 contains an electromotive force (usually a battery or generator) that produces a voltage of $E(t)$ volts (V) and a current of $I(t)$ amperes (A) at time t . The circuit also contains a resistor with a resistance of R ohm Ω and an inductor with an inductance of L henries (H). Ohm's Law gives the drop in voltage due to the resistor as RI . The voltage drop due to the inductor is $L\frac{dI}{dt}$. One of Kirchhoff's says that the sum of voltage drops is equal to the supplied voltage $E(t)$. Thus we have

$$L\frac{dI}{dt} + RI = E(t)$$

which is a first order linear differential equation. The solution gives the current I at time t .

- (a) Suppose that in the simple circuit of Figure 1, the resistance is 12Ω and the inductance is 4H . If a battery gives a constant voltage of 60V and the switch is closed when $t = 0$, so the current starts with $I(0) = 0$, find
- i. $I(t)$
 - ii. the current after 1 sec

ANSWER:

- 1 (a) $2y + \sin 2y = 4 \arctan x + C$.
 (c) $x = \frac{Ce^t}{Ce^t - 1}, x = 1$.
 (e) $y = \frac{1}{C - e^{\cos x}}$.
- 2 (a) $y = 2e^{\frac{-x^4}{4}} + 1$.
 (c) $y = \sin^2 x + 2 \sin x$.
 (d) $y = \arctan(1 + x^2)$.
 a. $r = \sin \theta + C \cos \theta$. b. $y = -t - 2 + Ce^t$.
- 3 c. $y = 1 + C(x^2 + 1)^{\frac{-1}{2}}$.
- 4 (b) $x = \frac{t^{-1}}{2} - 2t^{-3}$.